

Unit 8 – Inference for Categorical Data: Chi-Square

Notes 1 – The Chi-Square Goodness of Fit Test

Expected Counts

- A goodness-of-fit test is used to test the hypothesis that an observed frequency distribution fits to some claimed distribution
- An example of an equal expected count might be the following:
- A fair, six-sided die is rolled 60 times. What would be the expected count of each outcome?

Outcome	1	2	3	4	5	6
Expected Count						

We found the expected count by taking the total number of trials and dividing it equally among each of the outcomes.

- An example of an unequal expected count might be the following:
- An unfair, six-sided die is rolled 60 times. The die is loaded so that the number 1 turns up 50% of the time and the other five outcomes occur 10% of the time. What would be the expected count of each outcome?

Outcome	1	2	3	4	5	6
Probability						
Expected Count						

We found the expected count by taking the total number of trials and multiplying it by the probability of the outcome.

Observed Counts

While we know what is expected, when we run a simulation of rolling a fair die 60 times, we do not expect each outcome to be observed exactly 10 times each due to sampling variability.

Let's say I gave you a die I claimed was fair, and you observed the following counts when rolling the die 60 times:

Outcome	1	2	3	4	5	6
Expected Count	10	10	10	10	10	10
Observed Count	16	7	15	10	4	8

- Do you suspect that I did not give you a fair die?
- This is going to be the question our new test helps us answer: *Are the differences between the actual observed counts and the expected counts significant?*

Note: The observed counts must be all whole numbers because they represent actual counts, but the expected counts do not need to be whole numbers.

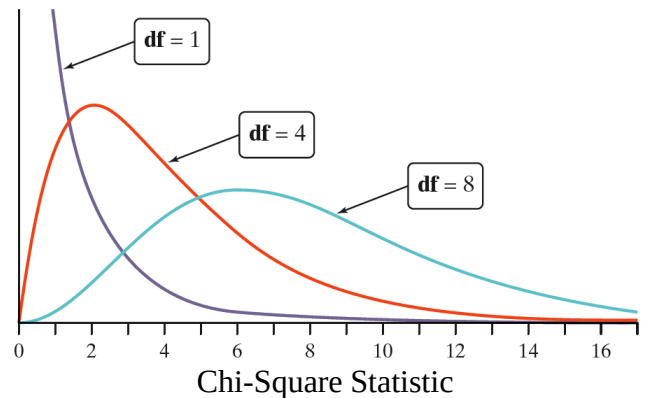
Measuring the Difference

To measure the difference between the observed and expected counts, and to determine if the difference is significant, we will introduce a new test statistic, called the chi-square statistic:

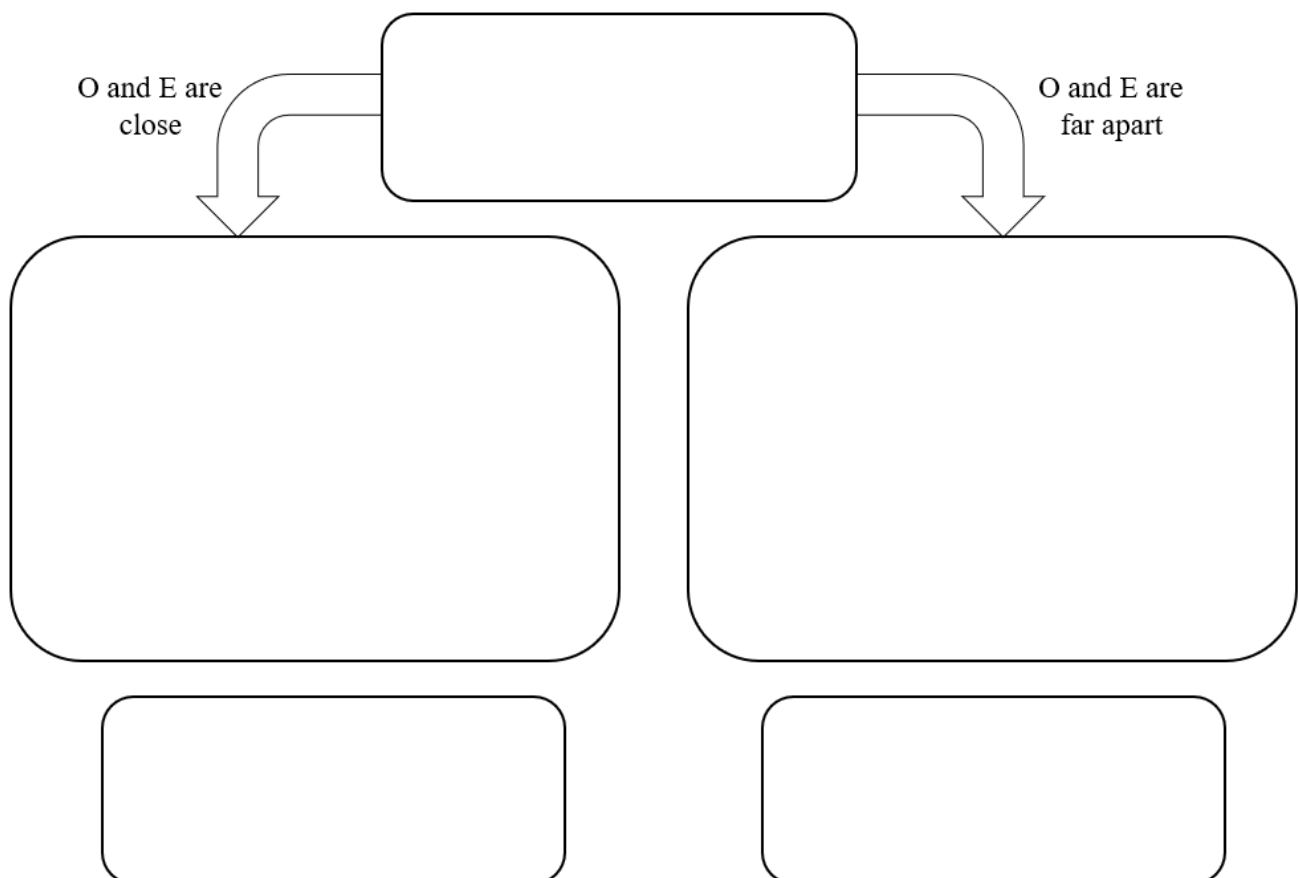
Where O represents each observed count in the distribution and E represents each corresponding expected count.

- The sampling distribution of the chi-square statistic is **NOT** a Normal distribution.
- It is a right-skewed distribution that allows only for positive values because the statistic cannot be negative.

When the expected counts are all at least 5, the sampling distribution of the χ^2 statistic is close to a chi-square distribution with degrees of freedom (df) equal to the number of categories minus 1.



- The chi-square distributions are a family of distributions that take only positive values and are skewed to the right.
- A particular chi-square distribution is specified by giving its degrees of freedom.
- The chi-square goodness-of-fit test uses the chi-square distribution with df = _____



Hypotheses

- The null hypothesis in a chi-square goodness-of-fit test should state a claim about the distribution of a single categorical variable in the population of interest.
- We can write this in words or symbols; both are acceptable and used on the AP Exam.
- Using our fair die as an example, we could say:

H_0 :	H_0 :
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The alternative hypothesis in a chi-square goodness-of-fit test is that the categorical variable does not have the specified distribution, and is easily given in words:

H_a : _____

Conditions

Random

- The data came from a well-designed random sample or randomized experiment.

Independent

- When sampling without replacement, the 10% condition is met.

Large Counts

- All expected counts are at least 5
- This allows us to say that the sampling distribution will follow a Chi-Square distribution

Chi-Square GOF Test

When the conditions are met, the chi-square goodness of fit test can be performed with the hypotheses:

H_0 : The claimed distribution is correct.

H_a : At least one proportion in the claimed distribution is incorrect.

We find the expected counts assuming the claimed distribution is true, and then we calculate the chi-square statistic:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

The p-value is the area to the right of χ^2 under the density curve of the chi-square distribution with $k - 1$ degrees of freedom, where k represents the number of categories.

$$P\text{-value} = P(\chi^2 > \text{value}) = \chi^2 \text{cdf}(\text{value}, 1e99, \text{df}) \text{ on the TI-84}$$

BEWARE

1. The chi-square test statistic compares observed and expected counts. Don't try to perform calculations with the observed and expected *proportions* in each category.
2. When checking the Large Counts condition, be sure to examine the expected counts, not the observed counts.

Example

A geneticist is studying the gene pattern of eye color in a group of white mice. He observed a random sample of mice from the lab and found that 110 had red eyes, 57 had brown eyes, 32 had pink eyes, and 13 had blue eyes. His model suggests that this distribution of eye color should occur in a 9:3:3:1 ratio. Is there evidence that the geneticist's model is not accurate? Use a 5% level of significance.